Report

Problem Statement

Develop a C++ program modeling webpages as vertices and hyperlinks as directed edges. Apply DFS from a chosen page, showing visitation order, reachable pages, and unvisited pages. Modify it to accept a different starting vertex and count reachable vertices or display BFS levels.

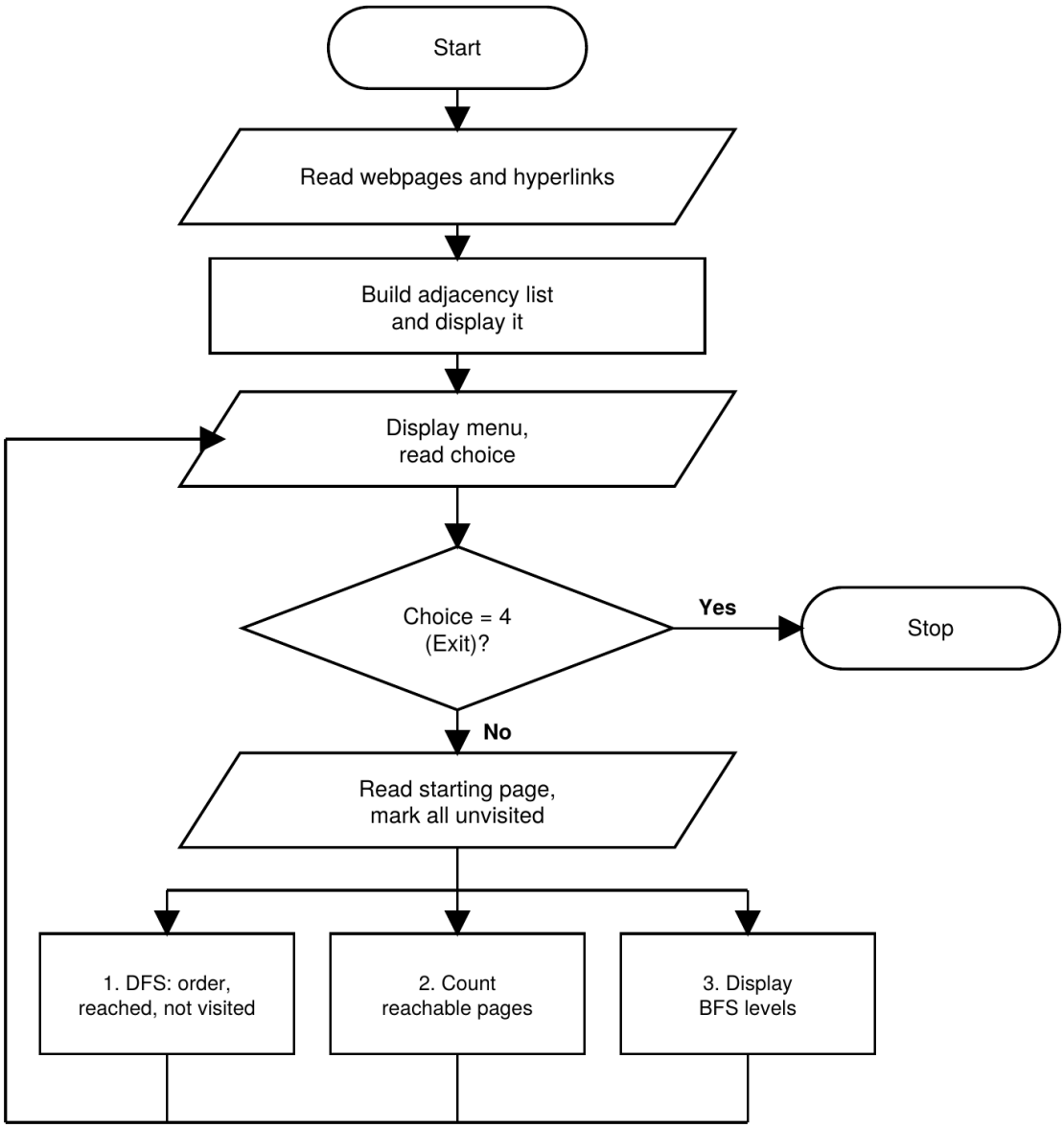
Objective

To implement a C++ program that models webpages as a directed graph, applies DFS from a user-selected starting page, displays the visitation order, reachable and unvisited pages, and counts reachable pages or displays their BFS levels.

Algorithm

1. Start.
2. Read the number of webpages and their names.
3. Read the hyperlinks and store each as a directed edge in the adjacency list.
4. Display the webpages with their hyperlinks.
5. Display the menu and read the choice.
6. Read the starting page and mark all pages unvisited.
7. For DFS, visit the page, store it in order, and recursively visit each unvisited linked page.
8. Display the visitation order, reached pages, and unvisited pages.
9. For count, display the number of reachable pages from the DFS result.
10. For BFS, use a queue to visit pages level by level and display each level.
11. Repeat the menu until Exit, then stop.

Flowchart


Start
Read webpages and hyperlinks
Build adjacency list
and display it
Display menu,
read choice
Choice = 4
(Exit)?
Yes
Stop
No
Read starting page,
mark all unvisited
1. DFS: order,
reached, not visited
2. Count
reachable pages
3. Display
BFS levels